



Original article

Examining the Bidirectional Associations Between Sleep Duration, Screen Time, and Internalizing Symptoms in the ABCD Study

Jennifer Zink, Ph.D.^{a,*}, Sydney G. O'Connor, Ph.D.^a, Dara R. Blachman-Demner, Ph.D.^b, Dana L. Wolff-Hughes, Ph.D.^c, and David Berrigan, Ph.D., M.P.H.^a^a Division of Cancer Control and Population Sciences, Health Behaviors Research Branch, Behavioral Research Program, National Cancer Institute, National Institutes of Health, Rockville, Maryland^b Office of Behavioral and Social Sciences Research, National Institutes of Health, Bethesda, Maryland^c Division of Cancer Control and Population Sciences, Risk Factor Assessment Branch, Epidemiology and Genomics Research Program, National Cancer Institute, National Institutes of Health, Rockville, Maryland

Article history: Received January 30, 2023; Accepted September 6, 2023

Keywords: Anxiety; Depression; Reciprocal; Sedentary behavior; Youth

A B S T R A C T

Purpose: The likelihood of meeting sleep duration and screen time guidelines decreases as children develop toward adolescence. Simultaneously, the prevalence of internalizing symptoms increases. The purpose of this paper was to examine the bidirectional associations between sleep duration and screen time with internalizing symptoms in a one-year longitudinal study starting in late childhood.

Methods: Participants were 10,828 youth (47.8% female) enrolled in the Adolescent Brain Cognitive Development Study. At baseline (mean age 9.9 years) and one-year follow-up (mean age 10.9 years), youth self-reported screen time for weekdays and weekend days. Responses were separately dichotomized as >2 versus ≤2 hours/day (meeting behavioral guidelines). Caregiver-reported youth sleep duration was dichotomized as <9 versus 9–11 hours/night (meeting behavioral guidelines). Caregivers reported internalizing symptoms via the child behavior checklist. The withdrawn/depressed, anxious/depressed, and somatic symptom child behavior checklist subscale t-scores were separately dichotomized as ≥65 (borderline clinical levels of symptoms and above) versus <65. Analyses were gender-stratified.

Results: In females, longer baseline sleep duration was protective against withdrawn/depressed symptoms (odds ratio [OR] 0.6, 95% confidence interval [CI] 0.4–0.8) and somatic complaints (OR 0.8, 95% CI 0.6–0.97) one year later. In females, greater baseline weekend screen time was associated with increased risk of withdrawn/depressed symptoms (OR 1.6, 95% CI 1.1–2.2) one year later. No other significant associations were observed.

Discussion: Longitudinal associations between sleep duration, weekend screen time, and internalizing symptoms were unidirectional (behavior preceding internalizing symptoms), among females only, and specific to withdrawn/depressed and somatic symptoms. These prospective study findings warrant attention and inform future research in this cohort.

Published by Elsevier Inc. on behalf of Society for Adolescent Health and Medicine.

IMPLICATIONS AND
CONTRIBUTION

As children develop toward adolescence, internalizing symptoms become increasingly prevalent. This study found that sleep and screen time longitudinally increased risk of internalizing symptoms immediately prior to this critical developmental period in females, setting the stage for further prospective and intervention studies of sleep, screen time, and internalizing symptoms in adolescents.

Conflicts of interest: The authors have no conflicts of interest to declare.

* Address correspondence to: Jennifer Zink, Ph.D., National Cancer Institute, 9609 Medical Center Drive, 3E530, Rockville, MD 20850.

E-mail address: jennifer.zink@nih.gov (J. Zink).

Internalizing symptoms are characterized by disordered mood and encompass depressive and anxiety symptoms [1]. These types of symptoms become more prevalent as children age into adolescence, particularly at the onset of puberty [2].

Significant associations between advancing pubertal stage and increased risk of internalizing symptoms are often observed, especially in females [3]. Accordingly, an estimated 25% and 20% of adolescents have significant levels of depressive and anxiety symptoms, respectively, with females being disproportionately affected [4]. Internalizing symptoms are a precursor for major depressive disorder and anxiety disorders, some of the most prevalent mental health conditions in youth, which have long-standing effects on development [2,5]. Even internalizing symptoms below the clinical threshold of these mental health conditions are associated with negative outcomes, including poorer social functioning, increased risk of substance use problems, and increased risk of overweight/obesity [2]. Therefore, reducing internalizing symptoms immediately before and during the child-to-adolescent transition may prevent poor outcomes across several domains.

Because of this, researchers have aimed to increase our understanding of potentially modifiable behavioral correlates of internalizing symptoms. It is recommended that youth sleep 9–12 hours/night [6] and limit leisure screen time to 2 hours/day [7], given mounting evidence of the health consequences of not meeting these guidelines. Paralleling the rise in internalizing symptoms across development, youth are simultaneously less likely to meet sleep duration and screen time recommendations [8]. For example, pubertal maturation triggers evening circadian preferences, with adolescents preferring later sleep/wake times compared to children [9]. This typically results in shorter sleep duration as youth age. An estimated ~89% and ~45% of children aged 6–11 years meet sleep duration and screen time recommendations, respectively [8]; by adolescence, the prevalence of each goes down to ~83% and ~21% [8]. According to separate studies, there is evidence that not meeting sleep duration and screen time recommendations confers some elevated risk of experiencing internalizing symptoms [10,11]. However, several gaps in our understanding of the associations between sleep duration, screen time, and internalizing symptoms remain.

Sleep duration and screen time are inversely related to one another, and only 30% of youth concurrently meet guidelines for both behaviors [8]. Relatively few studies simultaneously examine sleep duration and screen time in relation to internalizing symptoms. Therefore, the independent and interactive effects these behaviors may have on emotional outcomes are understudied, despite evidence from other fields suggesting that it is the combination of behaviors, rather than a single behavior alone, that may be most important for health outcomes [12]. Additionally, a recent systematic review highlights that a majority of the work that examines both behaviors in relation to internalizing symptoms is cross-sectional [13]. Of the limited longitudinal studies that have been conducted, one study of high school students found that females who met screen time and sleep duration guidelines had fewer depressive symptoms compared to females who did not [14]. Another longitudinal study during mid-adolescence that took a compositional analysis approach had similar gender-specific findings [15]. Although these studies were important for establishing prospective evidence and gender differences in these associations, they did not distinguish between different types of internalizing symptoms (e.g., depressive vs. anxiety), which could be differentially related to behaviors [16].

Another important consideration is possible reverse causality. For example, sleep disturbances are common features of internalizing symptoms, resulting in either excessive or inadequate

sleep duration [17]. Similarly, youth who experience internalizing symptoms may seek distraction from these emotional problems by turning to screen time [18]. Separate studies suggest that sleep duration and screen time are bidirectionally related to internalizing symptoms, but also point to nuances and complexities in these associations [11,19]. Therefore, more work is needed to increase our understanding of the possible longitudinal and bidirectional associations between sleep duration, screen time, and internalizing symptoms.

Our primary aims were to examine the independent and interactive associations between sleep duration, screen time, and internalizing symptoms in a one-year longitudinal study starting in late childhood, stratified by gender. We hypothesized that short sleep duration and excessive screen time (based on behavioral guidelines) would be independently related to an increased risk of internalizing symptoms one year later. We also hypothesized that these behaviors would interact and that the combination of short sleep duration and excessive screen time would render the greatest risk of internalizing symptoms one year later. Our secondary aim was to examine potential bidirectionality; we hypothesized that internalizing symptoms would also relate to short sleep duration and excessive screen time one year later. Lastly, we hypothesized that these associations would be stronger in females compared to males. Together, addressing these aims will move the field toward a causal understanding of the relationships between sleep duration, screen time, and internalizing symptoms.

Methods

Study design and sample

Data were from the Adolescent Brain Cognitive Development (ABCD) Study, a multi-site longitudinal study initiated in 2017 and coordinated by the United States (U.S.) National Institutes of Health [20]. Study participants are a large sample of youth ($N = 11,876$) aged 9–11 years at baseline, approximately representative of all U.S. youth [20]. Data is collected across 22 study sites on a biennial-to-biannual basis and is expected to run for at least 10 years [21]. Further details on study procedures are available elsewhere [21]. Ethics clearance was obtained from all relevant institutional review boards. Informed written consent and written assent were obtained from all parents/guardians and youth, respectively.

After clearance for data use was obtained, data were accessed through the national data archive. We used data from ABCD Release 4.0 (September 2021), containing baseline and one-year follow-up data for the full cohort collected before the COVID-19 pandemic. The analytic sample ($N = 10,828$) included participants with complete exposure and outcome data at both time points and complete baseline covariate data.

Measures

Sleep duration. Caregivers reported the child's sleep duration with an item from the Sleep Disturbance Scale for Children [22]. They were asked "How many hours of sleep does your child get on most nights?" with the following closed response options: 9–11; 8–9; 7–8; 5–7; less than 5 hours. It was not possible to report >11 hours. Responses were dichotomized as 9–11 hours versus <9 hours, consistent with the current guidelines [6].

Screen time. Youth self-reported their screen time via the ABCD Youth Screen Time Survey. This instrument asks participants to report their usual time spent (hours/day) on six different screen types (viewing/streaming television shows or movies, watching/streaming videos [e.g., YouTube], playing videogames, texting, video chatting, and on social networking sites) on weekdays and weekend days separately. Response options for each screen type included none, <30 minutes, 1 hour, 2 hours, 3 hours, and 4+ hours. Responses were summed to generate two separate screen time variables, total screen time on weekdays and total screen time on weekend days. Youth ($N = 243$) who reported >18 hours/day of total screen time on either weekdays or weekend days were removed prior to analysis [23]. Responses for weekdays and weekend days were treated separately since screen time and type differ between weekdays and weekend days [24]. Each screen time variable was dichotomized as >2 hours/day versus ≤ 2 hours/day, consistent with current guidelines [7].

Internalizing symptoms. Caregivers reported the youth's internalizing symptoms using the child behavior checklist (CBCL/4–18) [25], which is comprised of 113 items measuring behavioral and emotional problems in the past 6 months on a scale from 0 (never) to 2 (very true). Responses are used to generate eight subscale scores, three of which reflect internalizing symptoms; withdrawn/depressed (i.e., would rather be alone), anxious/depressed (i.e., worries), and somatic complaints (i.e., tired). Although some versions of the CBCL contain specific items about sleep (i.e., trouble getting to sleep), the internalizing symptom subscales used in the current study did not [25]. T-scores (mean of 50 and standard deviation [SD] of 10) for each of these three subscales were dichotomized as ≥ 65 (borderline clinical levels of symptoms and above/at or above the 93rd percentile) and <65 (within the normal range/below the 93rd percentile) [25]. Each subscale was examined separately, given their potentially unique associations with health behaviors [11,16].

Covariates. Selected covariates were consistent with prior work [26] and included baseline participant age (continuous; years), caregiver-reported child race/ethnicity (categorical; non-Hispanic Asian, non-Hispanic Black, Hispanic, Other [including Mixed Race], and non-Hispanic White), and socioeconomic status operationalized as the income-to-needs ratio [27]. Combined annual household income and household size for the income-to-needs ratio calculation were caregiver-reported. The income-to-needs ratio was then categorized as below the 2017 poverty threshold (≤ 0.99), low income (1.00–1.99), intermediate income (2.00–3.99), and high income (≥ 4.00) [27]. Participants missing annual household income ($N = 862$) or household size ($N = 146$) were categorized as 'not reported' and included in the analysis. Race/ethnicity and socioeconomic status were included as covariates based on evidence to suggest that health behaviors and internalizing symptoms can differ by these characteristics [2,28].

Covariates also included body mass index (kg/m^2), as weight status can be related to sleep duration, screen time, and internalizing symptoms [2,29]. Body mass index was calculated based on study staff-measured height and weight and was converted to percentiles using the 2000 Centers for Disease Control and Prevention growth charts. Percentiles were dichotomized as youth with overweight/obesity (≥ 85 th percentile) versus healthy weight/underweight (<85th percentile) [30]. Healthy weight

and underweight were combined due to few participants with underweight observed ($N = 41$). Youth-reported physical activity was assessed with the Youth Risk Behavior Surveillance System item "During the past 7 days, on how many days were you physically active for a total of at least 60 minutes per day? (Add up all the time you spent in any kind of physical activity that increased your heart rate and made you breathe hard at least some of the time)." Responses were treated continuously (days/week).

Statistical analysis

Descriptive statistics (mean [SD] or N [%]) were calculated for all study variables at baseline. For the sleep duration, screen time (on weekdays and weekend days), and internalizing symptom variables, N (%) was calculated at follow-up. McNemar's test was used to compare the prevalence of the sleep duration, screen time (on weekdays and weekend days), and internalizing symptom variables at baseline and follow-up. Bivariate correlations between all study variables at baseline were calculated prior to modeling; problematic multicollinearity was not observed.

To adjust for the clustering of observations within study sites, multilevel logistic regressions with the study site specified as a random effect were used to test the associations of interest using the SAS GLIMMIX procedure. First, three separate models with baseline sleep duration, and screen time (on weekdays and weekend days) as simultaneous predictors of each of the three internalizing symptom subscales at follow-up were run. To understand the potential interactive effects of baseline sleep duration and screen time (on weekdays or weekend days) on internalizing symptoms at follow-up, interaction terms were entered one at a time as predictors in each of the models and tested for significance. These terms were not included in subsequent models since significant interactions were not observed. Three further models addressing reverse directionality were run; the baseline withdrawn/depressed, anxious/depressed, and somatic complaints variables were entered as simultaneous predictors of each of the three behaviors at follow-up. All models were run unadjusted and adjusted for a priori covariates and each respective outcome variable at baseline.

All analyses were performed separately by gender a priori based on the participants' baseline gender identity, given consistent evidence to suggest that the associations of interest are stronger in females compared to males [14,15,31–33]. Gender identity was determined by caregiver responses to the following item, "What is the child's current gender identity?". Those who identified as male or transgender male ($N = 2$) were coded as male, and those who identified as female or transgender female ($N = 4$) were coded as female [31]. Participants who identified either as gender queer ($N = 2$) or different ($N = 4$) were not included in the analytic sample. Descriptive statistics of the main study variables were calculated for participants with these gender identities and are reported below. All analyses were conducted in SAS version 9.4.

Results

Description of the study sample at baseline

The analytic sample consisted of 10,828 participants (see Figure A1 for participant flow). The mean (SD) age of the analytic

sample was 9.9 (0.6) years at baseline and was 10.9 (0.6) years at follow-up. Just under half (45.7%) of the participants were of non-White racial/ethnic backgrounds with a wide range of socioeconomic status (income-to-needs ratio range: 0.1–15.4). Almost one-third (30.8%) of the sample had overweight/obesity and participants reported a mean (SD) of 3.5 (2.3) days/week of physical activity. Participant characteristics stratified by gender are reported in [Table 1](#).

Prevalence of behaviors and internalizing symptoms at baseline and one-year follow-up

At baseline, 48.8% of female participants reported 9–11 hours/night of sleep, and by one year later, this prevalence significantly decreased to 39.3% ($p < .0001$). The prevalence of >2 hours/day of screen time in female participants was 50.2% on weekdays and 65.5% on weekend days at baseline; both increased in prevalence (58.1% weekdays; 75.6% weekend days) by follow-up (p 's <0.0001). The prevalence of internalizing symptoms remained relatively stable across the study period, except for withdrawn/depressed symptoms, which increased from baseline (4.6%) to follow-up (5.5%) ($p = .01$) ([Table 2](#)).

Similar to female participants, approximately half (49.1%) of male participants reported 9–11 hours/night of sleep at baseline; by follow-up, this prevalence significantly decreased to 40.1% ($p < .0001$). The prevalence of >2 hours/day of screen time in male participants was 59.0% on weekdays and 76.5% on weekend days at baseline; both further increased in prevalence (66.6% weekdays; 84.2% weekend days) by follow-up (p 's <0.0001). Internalizing symptoms did not significantly change across the study period in males ([Table 2](#)).

The prevalence of behaviors and internalizing symptoms at baseline and one-year follow-up for participants identifying as gender queer ($N = 2$) and different genders ($N = 4$) is presented in [Table 2](#).

Table 1

Participant characteristics of the analytic sample and stratified by gender at baseline ($N = 10,828$)

	Mean (SD) or N (%)		
	All	Females ($N = 5,177$)	Males ($N = 5,651$)
Age (years), mean (SD)	9.9 (0.6)	9.9 (0.6)	9.9 (0.6)
Race/ethnicity, N (%)			
Non-Hispanic Asian	235 (2.2%)	121 (2.3%)	114 (2.0%)
Non-Hispanic Black	1,437 (13.3%)	722 (14.0%)	715 (12.7%)
Hispanic	2,149 (19.8%)	1,031 (19.9%)	1,118 (19.8%)
Other/mixed race	1,132 (10.4%)	550 (10.6%)	582 (10.3%)
Non-Hispanic White	5,875 (54.3%)	2,753 (53.2%)	3,122 (55.2%)
Income-to-needs ratio, N (%)			
≤0.99	1,292 (11.9%)	636 (12.3%)	656 (11.6%)
1.00–1.99	1,559 (14.4%)	737 (14.2%)	822 (14.5%)
2.00–3.99	2,484 (22.9%)	1,204 (23.3%)	1,280 (22.7%)
≥4.00	4,485 (41.4%)	2,141 (41.4%)	2,344 (41.5%)
Not reported	1,008 (9.3%)	459 (8.9%)	549 (9.7%)
Overweight/obesity, N (%)			
Physical activity (days/week), mean (SD)	3.5 (2.3)	3.4 (2.3)	3.6 (2.3)

The income-to-needs ratio was categorized as 'not reported' if caregiver responded 'don't know' or 'refuse to answer' to the combined annual household income survey item or if information on household size was not provided. SD = standard deviation.

Table 2

Prevalence (N [%]) of sleep duration, screen time, and internalizing symptoms at baseline and one-year follow-up stratified by gender

	Baseline	Follow-up	<i>p</i>
Females ($N = 5,177$)			
Sleep duration (9–11 hours/night)	2,528 (48.8%)	2,033 (39.3%)	<.0001
Screen time (>2 hours/day)			
Weekday	2,598 (50.2%)	3,008 (58.1%)	<.0001
Weekend	3,392 (65.5%)	3,916 (75.6%)	<.0001
Internalizing symptoms (t-score ≥65)			
Withdrawn/depressed	236 (4.6%)	282 (5.5%)	.01
Anxious/depressed	386 (7.5%)	376 (7.3%)	.61
Somatic complaints	448 (8.7%)	436 (8.4%)	.59
Males ($N = 5,651$)			
Sleep duration (9–11 hours/night)	2,774 (49.1%)	2,265 (40.1%)	<.0001
Screen time (>2 hours/day)			
Weekday	3,334 (59.0%)	3,766 (66.6%)	<.0001
Weekend	4,323 (76.5%)	4,758 (84.2%)	<.0001
Internalizing symptoms (t-score ≥65)			
Withdrawn/depressed	529 (9.4%)	550 (9.7%)	.37
Anxious/depressed	415 (7.3%)	412 (7.3%)	.88
Somatic complaints	417 (7.4%)	411 (7.3%)	.78
Gender queer ($N = 2$)			
Sleep duration (9–11 hours/night)	0 (0.0%)	0 (0.0%)	-
Screen time (>2 hours/day)			
Weekday	2 (100.0%)	2 (100.0%)	-
Weekend	2 (100.0%)	2 (100.0%)	-
Internalizing symptoms (t-score ≥65)			
Withdrawn/depressed	1 (50.0%)	2 (100.0%)	-
Anxious/depressed	2 (100.0%)	2 (100.0%)	-
Somatic complaints	1 (50.0%)	2 (100.0%)	-
Different gender Identity ($N = 4$)			
Sleep duration (9–11 hours/night)	1 (25.0%)	1 (25.0%)	-
Screen time (>2 hours/day)			
Weekday	0 (0.0%)	0 (0.0%)	-
Weekend	1 (25.0%)	4 (100.0%)	-
Internalizing symptoms (t-score ≥65)			
Withdrawn/depressed	1 (25.0%)	1 (25.0%)	-
Anxious/depressed	1 (25.0%)	2 (50.0%)	-
Somatic complaints	0 (0.0%)	0 (0.0%)	-

p value derived from McNemar's test.

Longitudinal associations between baseline behaviors and internalizing symptoms at one-year follow-up

Sleep duration. In females, 9–11 hours/night of sleep (vs. <9 hours) at baseline was protective against subsequent withdrawn/depressed symptoms (adjusted odds ratio [OR] 0.6, 95% confidence interval [CI] 0.4–0.8, $p = .0007$) and somatic complaints (adjusted OR 0.8, 95% CI 0.6–0.97, $p = .0275$), but was unrelated to subsequent anxious/depressed symptoms. In males, baseline sleep duration was unrelated to subsequent withdrawn/depressed symptoms, anxious/depressed symptoms, and somatic complaints ([Table 3](#)).

Screen time. In females and males, >2 hours/day of screen time on weekdays (vs. ≤2 hours/day) was unrelated to subsequent withdrawn/depressed symptoms, anxious/depressed symptoms, and somatic complaints ([Table 3](#)). In females, >2 hours/day of

Table 3

Estimates of the association (OR [95% CI]) between baseline sleep duration and screen time and internalizing symptoms at one-year follow-up stratified by gender (N = 10,828)

	Withdrawn/depressed (t-score ≥ 65)		Anxious/depressed (t-score ≥ 65)		Somatic complaints (t-score ≥ 65)	
	Unadjusted	Adjusted	Unadjusted	Adjusted	Unadjusted	Adjusted
Females (N = 5,177)						
Sleep duration						
<9 hours/night	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
9–11 hours/night	0.6 (0.4–0.7)****	0.6 (0.4–0.8)***	0.8 (0.6–1.1)	0.8 (0.7–1.0)	0.7 (0.6–0.9)**	0.8 (0.6–0.97)*
Screen time (weekday)						
≤ 2 hours/day	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
>2 hours/day	1.1 (0.8–1.4)	0.7 (0.5–1.0)	1.1 (0.9–1.4)	1.0 (0.7–1.3)	1.3 (1.0–1.6)*	1.0 (0.8–1.3)
Screen time (weekend)						
≤ 2 hours/day	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
>2 hours/day	1.5 (1.1–2.0)**	1.6 (1.1–2.2)*	1.2 (0.9–1.5)	1.1 (0.8–1.5)	1.4 (1.1–1.8)**	1.3 (1.0–1.7)
Males (N = 5,651)						
Sleep duration						
<9 hours/night	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
9–11 hours/night	0.8 (0.7–1.0)	1.0 (0.8–1.2)	1.0 (0.8–1.3)	1.0 (0.8–1.3)	0.7 (0.6–0.9)**	0.8 (0.6–1.0)
Screen time (weekday)						
≤ 2 hours/day	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
>2 hours/day	1.2 (1.0–1.4)	1.1 (0.8–1.4)	1.1 (0.9–1.4)	1.1 (0.9–1.5)	1.3 (1.0–1.6)*	1.0 (0.8–1.3)
Screen time (weekend)						
≤ 2 hours/day	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
>2 hours/day	1.0 (0.8–1.3)	0.9 (0.7–1.2)	1.0 (0.7–1.3)	0.9 (0.7–1.2)	1.4 (1.0–1.8)*	1.3 (0.9–1.8)

Unadjusted models include either sleep duration, screen time (weekday), or screen time (weekend day) and the respective outcome at baseline as the only two predictors. Adjusted models include sleep duration, screen time (weekday), and screen time (weekend day) as simultaneous predictors, adjusting for a priori covariates and the respective outcome at baseline.

OR = odds ratio; CI = confidence interval.

* $p < .05$ ** $p < .01$ *** $p < .001$ **** $p < .0001$.

screen time on weekend days (vs. ≤ 2 hours/day) at baseline was a risk factor for subsequent withdrawn/depressed symptoms (adjusted OR 1.6, 95% CI 1.1–2.2, $p = .0129$). However, screen time on weekend days at baseline was unrelated to subsequent

anxious/depressed symptoms and somatic complaints in females. It was also unrelated to subsequent withdrawn/depressed symptoms, anxious/depressed symptoms, and somatic complaints in males (Table 3).

Table 4

Estimates of the association (OR [95% CI]) between baseline internalizing symptoms and sleep duration and screen time at one-year follow-up stratified by gender (N = 10,828)

	Sleep duration (9–11 hours/night)		Screen time (weekday) (> 2 hours/day)		Screen time (weekend) (> 2 hours/day)	
	Unadjusted	Adjusted	Unadjusted	Adjusted	Unadjusted	Adjusted
Females (N = 5,177)						
Withdrawn/depressed						
t-score < 65	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
t-score ≥ 65	0.7 (0.5–0.9)*	0.7 (0.5–1.1)	1.1 (0.8–1.5)	0.8 (0.6–1.2)	1.1 (0.8–1.5)	0.9 (0.6–1.3)
Anxious/depressed						
t-score < 65	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
t-score ≥ 65	0.8 (0.6–1.0)	0.9 (0.7–1.2)	1.3 (1.0–1.6)*	1.2 (1.0–1.6)	1.2 (0.9–1.5)	1.1 (0.8–1.5)
Somatic complaints						
t-score < 65	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
t-score ≥ 65	0.8 (0.7–1.1)	0.9 (0.7–1.2)	1.3 (1.0–1.6)*	1.2 (0.9–1.5)	1.2 (0.9–1.5)	1.1 (0.9–1.5)
Males (N = 5,651)						
Withdrawn/depressed						
t-score < 65	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
t-score ≥ 65	0.7 (0.6–0.9)**	0.9 (0.7–1.1)	1.1 (0.9–1.4)	1.0 (0.8–1.3)	1.0 (0.8–1.3)	1.0 (0.7–1.3)
Anxious/depressed						
t-score < 65	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
t-score ≥ 65	0.8 (0.7–1.1)	1.0 (0.7–1.3)	1.0 (0.8–1.3)	1.0 (0.7–1.3)	0.9 (0.7–1.3)	0.9 (0.6–1.2)
Somatic Complaints						
t-score < 65	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)	1.0 (ref.)
t-score ≥ 65	0.8 (0.6–0.97)*	0.9 (0.7–1.1)	1.0 (0.8–1.3)	0.9 (0.7–1.2)	1.0 (0.8–1.4)	1.0 (0.7–1.4)

Unadjusted models include either withdrawn/depressed, anxious/depressed, or somatic complaints and the respective outcome at baseline as the only two predictors. Adjusted models include withdrawn/depressed, anxious/depressed, and somatic complaints as simultaneous predictors, adjusting for a priori covariates and the respective outcome at baseline.

OR = odds ratio; CI = confidence interval.

* $p < .05$ ** $p < .01$.

Longitudinal associations between baseline internalizing symptoms and behaviors at one-year follow-up

Withdrawn/depressed. In females and males, being at or above borderline clinical levels of withdrawn/depressed symptoms (vs. not) at baseline was unrelated to subsequent sleep duration and screen time (on weekdays and weekend days) (Table 4).

Anxious/depressed. Similarly, being at or above borderline clinical levels of anxious/depressed symptoms (vs. not) at baseline was unrelated to subsequent sleep duration and screen time (on weekdays and weekend days) in females and males (Table 4).

Somatic complaints. Lastly, being at or above borderline clinical levels of somatic complaints (vs. not) at baseline was unrelated to subsequent sleep duration and screen time (on weekdays and weekend days) in females and males (Table 4).

Sensitivity analyses

Sensitivity analyses were also conducted using a survey design-based approach with the SAS SURVEYLOGISTIC procedure, using the study site as the cluster variable and sample weights applied to approximate the American Community Survey [34]. Results were consistent with the random effects models (Tables A1 and A2). Sensitivity analyses using baseline pubertal status ($N = 8,716$) as a covariate instead of baseline age were also conducted; results were similar to those presented above (Tables A3 and A4).

Although we were interested in understanding these relationships in the context of borderline (and above) clinical levels of internalizing symptoms (dichotomized), we additionally examined the bidirectional associations between continuous CBCL t-scores, sleep duration, and screen time (Tables A5 and A6). Significant associations between baseline behavior and continuous CBCL t-scores at follow-up were not observed in females. In males, >2 hours/day of screen time on weekdays (vs. ≤ 2 hours/day) at baseline was associated with higher withdrawn/depressed and anxious/depressed t-scores one year later. In females, higher somatic complaints t-scores at baseline were a risk factor for >2 hours/day (vs. ≤ 2 hours/day) screen time on weekdays and weekend days at one-year follow-up. Significant associations between baseline continuous CBCL t-scores and behavior at follow-up were not observed in males. Together, this suggests that these relationships may differ depending on whether internalizing symptoms at/above borderline clinical levels (dichotomized) are being examined or whether any level of internalizing symptoms regardless of clinical significance (continuous) are being examined.

Discussion

We investigated the longitudinal and bidirectional associations between sleep duration, screen time (separately on weekdays and weekend days), and internalizing symptoms across one year in a large cohort of U.S. youth aged 10 years at baseline. Our results indicate that sleep duration and screen time are independently and longitudinally associated with internalizing symptoms in females only, and that these associations are unidirectional. These findings provide important prospective evidence that should be explored further in future work. Our gender-specific findings are consistent with previous work

[14,15,31–33], and our longitudinal findings extend prior cross-sectional analyses that identified associations between sleep duration, screen time, and internalizing symptoms/diagnoses in the ABCD sample [26,31]. We also extend prior longitudinal work in the ABCD sample by suggesting that these associations may be specific to depressive and somatic symptoms and by exploring potential bidirectionality [35]. The gender- and symptom-specificity of the associations that we observed warrants attention in future work, particularly as this cohort ages.

The primary aim of this study was to determine whether sleep duration and screen time were independently and/or interactively related to internalizing symptoms one year later, given that few longitudinal studies have simultaneously examined both behaviors in relation to emotional outcomes. We did not observe an interaction between sleep duration and screen time. This is consistent with prior cross-sectional work in children and adolescents [36,37]. Although these behaviors may not interact, it is possible that short sleep duration partially mediates the association between screen time and internalizing symptoms. Past studies have provided mixed evidence for mediation [38,39], but have suggested that short sleep duration may specifically mediate the relationship between nighttime screen use and internalizing symptoms [40]. It is possible that social jetlag, which reflects misalignment in sleep timing between days, is related to internalizing symptom risk in youth [33]; however, sleep timing was not assessed in the ABCD Study. More granular data on sleep, screen time, and their timing could help clarify the relationships at hand.

Notably, we observed associations specifically for weekend screen time. A recent systematic review suggests that the strength of the relationship between screen time and internalizing symptoms may depend on screen type [32]. Screen type-specific associations with internalizing symptoms could be attributed to differences in the content accessed across screen types/devices, such as whether there are opportunities for upward social comparison [41] or reinforcing spirals [42] to occur. It is also possible that certain screen types could promote social isolation [43]. In the current study, participants reported more total screen time on weekend days (compared to weekdays), which was specifically driven by more time spent viewing/streaming television shows/movies, watching/streaming videos (e.g., YouTube), and playing videogames (ancillary analyses, data not shown). This is consistent with previous reports in the ABCD sample [16,24]. To further understand the observed associations between weekend screen time and withdrawn/depressed symptoms in females, we conducted exploratory analyses to estimate the associations between different screen time categories (weekends only) and withdrawn/depressed symptoms (Table A7). Specific screen time categories on weekends were not associated with withdrawn/depressed symptoms. This is consistent with the possibility that weekend screen time was related to internalizing symptoms because participants spent more total time on screens during the weekends, rather than because they spent their time on different types of screens during the weekends. Further efforts to understand why weekend screen time may specifically relate to internalizing symptoms are important for informing intervention development and implementation.

Contrary to our hypothesis, internalizing symptoms were not related to sleep duration or screen time one year later in this sample. Prior investigations suggest reverse causality is plausible. A previous study examining the associations between sleep

problems (including short sleep duration) and internalizing symptoms across 15 years from early adolescence to young adulthood had similar findings to ours; sleep problems preceded internalizing symptoms (but not vice versa) during early- and mid-adolescence [44]. Another study in youth aged 13 years (baseline) to 17 years (follow-up) found that short sleep duration was related to subsequent emotional symptoms (but not vice versa) specifically between the ages of 13–15 years old [19]. The authors also noted a possible developmental process whereby these emotional symptoms at 15 years old, in turn, prospectively increased the risk of sleep difficulties (but not sleep duration) at 17 years old [19]. Similar to the complexities noted in the sleep literature, it is also possible that internalizing symptoms may precede only specific screen types, and that these associations likely emerge in older youth than studied here [11]. Taken together, the directionality of these associations may not be stable across development, and other metrics of sleep (such as sleep problems/difficulties) and specific screen types are worth considering in future investigations. The current work sets the stage for longer-term investigations of these relationships across development through adolescence, and the ABCD Study will be an important data source for aiding these efforts moving forward.

Strengths of the current include the large, population-based sample and the longitudinal design. The simultaneous examination of sleep duration and screen time (and their possible interaction), the investigation of different types of internalizing symptoms separately, and the consideration of reverse causality are significant strengths. There are also limitations. Youth sleep duration and internalizing symptoms were caregiver-reported, despite youth possibly being better reporters than their caregivers by this age. Additionally, while the screen time survey differentiated between weekday and weekend day behaviors, the sleep questionnaire did not. Weekday-to-weekend day discrepancies in sleep duration and timing are common in school-aged youth [45], and future work could consider using device-based measures of sleep across several days to address these sleep measurement limitations. This could also capture additional sleep parameters, such as social jetlag, which may be important for internalizing symptom risk in youth [33]. Further, while physical activity was a covariate, it was not one of the primary behaviors of interest because it was measured at baseline only. This precluded our ability to examine reverse causality (our secondary aim). Future work could consider examining bidirectional associations between behaviors over 24 hours (sleep, screen time, and physical activity) and internalizing symptoms. Lastly, we did not adjust for pubertal status in the primary analysis, given the large number ($N = 2,112$) of participants with missing baseline pubertal data, particularly in females. Our sensitivity analyses adjusting for pubertal status yielded qualitatively similar results. However, as the ABCD cohort ages and a wider range of pubertal development is observed, this will be an important covariate and moderator worth exploring.

Conclusions

We investigated the independent and interactive associations between sleep duration, screen time, and internalizing symptoms in a one-year longitudinal study of the ABCD cohort. Sleep duration and screen time (on weekends only) at baseline were independently related to specific internalizing symptom dimensions (withdrawn/depressed and somatic) one year later.

These associations were observed in females only, and we did not find evidence of an interaction between sleep duration and screen time or evidence of reverse causality (internalizing symptoms preceding behavior). These findings inform future longitudinal studies in the ABCD cohort aimed at understanding how the complex relationships between sleep duration, screen time, and internalizing symptoms may unfold across development through adolescence.

Acknowledgments

Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive Development (ABCD) Study (<https://abcdstudy.org>), held in the NIMH data archive (NDA). This is a multisite, longitudinal study designed to recruit more than 10,000 children aged 9–10 and follow them over 10 years into early adulthood.

Funding Sources

The ABCD study is supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018, U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089, U24DA041123, U24DA041147. A full list of supports is available at <https://abcdstudy.org/federal-partners.html>. A list of participating sites and a complete list of the study investigators can be found at https://abcdstudy.org/consortium_members/. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report.

Supplementary Data

Supplementary data related to this article can be found online at <https://doi.org/10.1016/j.jadohealth.2023.09.001>.

References

- [1] Achenbach TM, McConaughy SH. Taxonomy of internalizing disorders of childhood and adolescence. *Internalizing disorders in children and adolescents*. Oxford, England: John Wiley & Sons; 1992:19–60.
- [2] Thapar A, Eyre O, Patel V, et al. Depression in young people. *Lancet* 2022; 400:617–31.
- [3] Stumper A, Alloy LB. Associations between pubertal stage and depression: A systematic review of the literature. *Child Psychiatry Hum Dev* 2021;54: 1–28.
- [4] Racine N, McArthur BA, Cooke JE, et al. Global prevalence of depressive and anxiety symptoms in children and adolescents during COVID-19: A meta-analysis. *JAMA Pediatr* 2021;175:1142–50.
- [5] Wehry AM, Beesdo-Baum K, Hennelly MM, et al. Assessment and treatment of anxiety disorders in children and adolescents. *Curr Psychiatry Rep* 2015; 17:1–11.
- [6] Paruthi S, Brooks LJ, D'Ambrosio C, et al. Consensus statement of the American Academy of sleep medicine on the recommended amount of sleep for healthy children: Methodology and discussion. *J Clin Sleep Med* 2016;12:1549–61.
- [7] Hill D, Ameenuddin N, Chassiakos YLR, et al. Media use in school-aged children and adolescents. *Pediatrics* 2016;138:e20162592.
- [8] Friel CP, Duran AT, Shechter A, et al. US children meeting physical activity, screen time, and sleep guidelines. *Am J Prev Med* 2020;59:513–21.
- [9] Carskadon MA, Vieira C, Acebo C. Association between puberty and delayed phase preference. *Sleep* 1993;16:258–62.

- [10] Chaput JP, Gray CE, Poitras VJ, et al. Systematic review of the relationships between sleep duration and health indicators in school-aged children and youth. *Appl Physiol Nutr Metab* 2016;41:S266–82.
- [11] Zink J, Ebrahimian S, Belcher BR, et al. Reciprocal associations between depression and screen-based sedentary behaviors in adolescents differ by depressive symptom dimension and screen-type. *J Affect Disord* 2020;263:39–46.
- [12] Chaput JP, Carson V, Gray CE, et al. Importance of all movement behaviors in a 24 hour period for overall health. *Int J Environ Res Public Health* 2014;11:12575–81.
- [13] Sampasa-Kanyinga H, Colman I, Goldfield GS, et al. Combinations of physical activity, sedentary time, and sleep duration and their associations with depressive symptoms and other mental health problems in children and adolescents: A systematic review. *Int J Behav Nutr Phys Act* 2020;17:1–16.
- [14] Patte KA, Faulkner G, Qian W, et al. Are one-year changes in adherence to the 24-hour movement guidelines associated with depressive symptoms among youth? *BMC Public Health* 2020;20:793.
- [15] Sampasa-Kanyinga H, Colman I, Dumuid D, et al. Longitudinal association between movement behaviours and depressive symptoms among adolescents using compositional data analysis. *PLoS One* 2021;16:e0256867.
- [16] Fors PQ, Barch DM. Differential relationships of child anxiety and depression to child report and parent report of electronic media use. *Child Psychiatry Hum Dev* 2019;50:907–17.
- [17] Franzen PL, Buysse DJ. Sleep disturbances and depression: Risk relationships for subsequent depression and therapeutic implications. *Dialogues Clin Neurosci* 2022;10:473–81.
- [18] Hoge E, Bickham D, Cantor J. Digital media, anxiety, and depression in children. *Pediatrics* 2017;140:S76.
- [19] Kortesoja L, Vainikainen MP, Hotulainen R, et al. Bidirectional relationship of sleep with emotional and behavioral difficulties: A five-year follow-up of Finnish adolescents. *J Youth Adolesc* 2020;49:1277–91.
- [20] Garavan H, Bartsch H, Conway K, et al. Recruiting the ABCD sample: Design considerations and procedures. *Dev Cogn Neurosci* 2018;32:16–22.
- [21] Barch DM, Albaugh MD, Avenevoli S, et al. Demographic, physical and mental health assessments in the adolescent brain and cognitive development study: Rationale and description. *Dev Cogn Neurosci* 2018;32:55–66.
- [22] Bruni O, Ottaviano S, Guidetti V, et al. The Sleep Disturbance Scale for Children (SDSC) Construct ion and validation of an instrument to evaluate sleep disturbances in childhood and adolescence. *J Sleep Res* 1996;5:251–61.
- [23] Clark BK, Lynch BM, Winkler EA, et al. Validity of a multi-context sitting questionnaire across demographically diverse population groups: AusDiab3. *Int J Behav Nutr Phys Act* 2015;12:1–9.
- [24] Paulich KN, Ross JM, Lessem JM, et al. Screen time and early adolescent mental health, academic, and social outcomes in 9-and 10-year old children: Utilizing the Adolescent Brain Cognitive Development™ (ABCD) Study. *PLoS One* 2021;16:e0256591.
- [25] Achenbach TM, Ruffle TM. The Child Behavior Checklist and related forms for assessing behavioral/emotional problems and competencies. *Pediatr Rev* 2000;21:265–71.
- [26] Sampasa-Kanyinga H, Colman I, Goldfield GS, et al. 24-hour movement behaviors and internalizing and externalizing behaviors among youth. *J Adolesc Health* 2021;68:969–77.
- [27] White SF, Voss JL, Chiang JJ, et al. Exposure to violence and low family income are associated with heightened amygdala responsiveness to threat among adolescents. *Dev Cogn Neurosci* 2019;40:100709.
- [28] Knell G, Durand CP, Kohl HW, et al. Prevalence and likelihood of meeting sleep, physical activity, and screen-time guidelines among US youth. *JAMA Pediatr* 2019;173:387–9.
- [29] Laurson KR, Lee JA, Gentile DA, et al. Concurrent associations between physical activity, screen time, and sleep duration with childhood obesity. *ISRN Obes* 2014;2014:204540.
- [30] Krebs NF, Himes JH, Jacobson D, et al. Assessment of child and adolescent overweight and obesity. *Pediatrics* 2007;120:S193–228.
- [31] Robertson L, Twenge JM, Joiner TE, et al. Associations between screen time and internalizing disorder diagnoses among 9-to 10-year-olds. *J Affect Disord* 2022;311:530–7.
- [32] Zink J, Belcher BR, Imm K, et al. The relationship between screen-based sedentary behaviors and symptoms of depression and anxiety in youth: A systematic review of moderating variables. *BMC Public Health* 2020;20:1–37.
- [33] Mathew GM, Hale L, Chang A-M. Sex moderates relationships among school night sleep duration, social jetlag, and depressive symptoms in adolescents. *J Biol Rhythms* 2019;34:205–17.
- [34] Heeringa SG, Berglund PA. A guide for population-based analysis of the Adolescent Brain Cognitive Development (ABCD) Study baseline data. *BioRxiv* 2020.
- [35] Fung H, Yeo BT, Chen C, et al. Adherence to 24-hour movement recommendations and health indicators in early adolescence: Cross-sectional and longitudinal associations in the adolescent brain cognitive development study. *J Adolesc Health* 2023;72:460–70.
- [36] Kahn M, Schnabel O, Gradsar M, et al. Sleep, screen time and behaviour problems in preschool children: An actigraphy study. *Eur Child Adolesc Psychiatry* 2021;30:1793–802.
- [37] Do YK, Shin E, Bautista MA, et al. The associations between self-reported sleep duration and adolescent health outcomes: What is the role of time spent on internet use? *Sleep Med* 2013;14:195–200.
- [38] Leung CY, Torres R. Sleep duration does not mediate the association between screen time and adolescent depression and anxiety: Findings from the 2018 national survey of children's health. *Sleep Med* 2021;81:227–34.
- [39] Guerrero MD, Barnes JD, Chaput JP, et al. Screen time and problem behaviors in children: Exploring the mediating role of sleep duration. *Int J Behav Nutr Phys Act* 2019;16:1–10.
- [40] Lemola S, Perkinson-Gloor N, Brand S, et al. Adolescents' electronic media use at night, sleep disturbance, and depressive symptoms in the smartphone age. *J Youth Adolesc* 2015;44:405–18.
- [41] Verduyn P, Gugushvili N, Massar K, et al. Social comparison on social networking sites. *Curr Opin Psychol* 2020;36:32–7.
- [42] Slater MD. Reinforcing spirals: The mutual influence of media selectivity and media effects and their impact on individual behavior and social identity. *Commun Theor* 2007;17:281–303.
- [43] Werneck AO, Collings PJ, Barboza LL, et al. Associations of sedentary behaviors and physical activity with social isolation in 100,839 school students: The Brazilian Scholar Health Survey. *Gen Hosp Psychiatry* 2019;59:7–13.
- [44] Narmandakh A, Roest AM, de Jonge P, et al. The bidirectional association between sleep problems and anxiety symptoms in adolescents: A TRAILS report. *Sleep Med* 2020;67:39–46.
- [45] Bei B, Allen NB, Nicholas CL, et al. Actigraphy-assessed sleep during school and vacation periods: A naturalistic study of restricted and extended sleep opportunities in adolescents. *J Sleep Res* 2014;23:107–17.